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Place a box around all final answers. Upload by 11:59 pm.

EE133 Solid-State Electronics, Midterm

NAME _____

Show all work for full credit. For all problems use:

For Si, $n_i = 10^{10} / \text{cm}^3$ at $T=300\text{K}$, $E_G = 1.12 \text{ eV}$ and $\epsilon_s = 11.9\epsilon_0$.

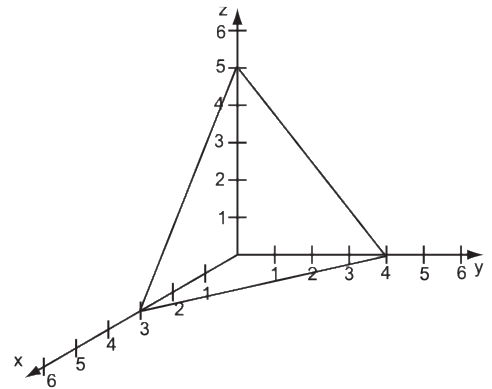
$kT = 0.026 \text{ eV}$ at $T=300\text{K}$, $q=1.602 \times 10^{-19} \text{ C}$, $\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$, $m_0 = 9.11 \times 10^{-31} \text{ kg}$.

1. (a) (10pts) What are the Miller indices of the x-y plane?

(b) (10pts) List all of the planes equivalent to the x-y plane.

(c) (10pts) Use the correct notation to indicate this full set of equivalent planes.

2. (10pts) What are the Miller indices of the plane at right



3. Si is B doped at a concentration of $10^{17} / \text{cm}^3$. $T=300\text{K}$.

a. (10 pts) Determine the electron density.

b. (10 pts) Determine the hole density.

c. (10 pts) Which best describes the material, (i) intrinsic (ii) extrinsic? Why?

d. (20 pts) What is the value (in eV) of $(E_i - E_F)$?

4. (10 pts) At room temperature, which semiconductor has the largest intrinsic carrier concentration? Explain based on an equation.
- Si ($E_G = 1.1$ eV)
 - Ge ($E_G = 0.7$ eV)
 - GaAs ($E_G = 1.4$ eV)
 - InAs ($E_G = 0.35$ eV)
 - InSb ($E_G = 0.17$ eV)
5. (10 pts) At room temperature, which semiconductor has the highest electron mobility? Explain.
- Si ($N_D = 10^{14}$ / cm³)
 - Si ($N_D = 10^{17}$ / cm³)
 - Si ($N_D = 10^{18}$ / cm³)
 - Both (a) and (b) have same mobility.
6. (10 pts) Which of the above in question 5 has the highest conductivity? Explain based on an equation.

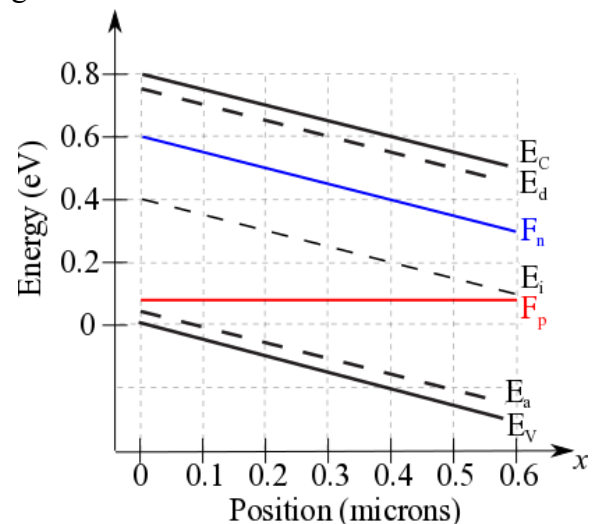
7. Answer the following questions based on the band diagram at right.

(a) (5pts) It is in equilibrium. T F

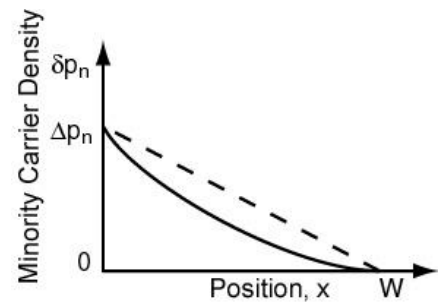
(b) (5 pts) The electric field is pointing in which direction:
 (i) Left (ii) Right (iii) Neither since it is zero.

(c) (5pts) There is an electron drift current flowing, (i) to the left, (ii) to the right, (iii) neither since it is zero.

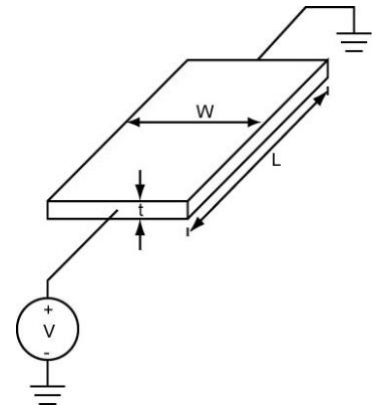
(d) (5pts) There is an electron diffusion current flowing (i) to the left, (ii) to the right, (iii) neither since it is zero.



- (e) (5pts) There is a hole drift current flowing (i) to the left, (ii) to the right, (iii) neither since it is zero.
- (f) (5pts) There is a hole diffusion current flowing (i) to the left, (ii) to the right, (iii) neither since it is zero.
- (g) (5pts) The total electron current is flowing (i) to the left, (ii) to the right, (iii) neither since it is zero.
- (h) (5pts) The total hole current is flowing (i) to the left, (ii) to the right, (iii) neither since it is zero.
8. The figure at right shows two different minority carrier hole distributions.
- (a) (5pts) Circle the one which results in the largest diffusion current at $x=0$.
- (b) (5pts) Explain using an equation.
- (c) (5pts) What direction is the hole diffusion current flowing?
- (d) (10pts) Comparing the distribution of the solid curve to the dashed curve, what physical mechanism is present or absent that gives rise to the 2 different curves? (Consider the 4 solutions of the MCDEs).



9. A bar of intrinsic Si at $T=300\text{K}$ is shown at right. The dimensions of the bar are $L = 10\text{ }\mu\text{m}$, $W = 1\text{ }\mu\text{m}$, and $t = 0.1\text{ }\mu\text{m}$. It is doped p-type with $N_A = 10^{18}\text{ / cm}^3$. The mobility is $\mu_p = 642.2\text{ cm}^2/\text{Vs}$. $m_p^* = 0.4m_0$. The current density is $J = 1 \times 10^5\text{ A/cm}^2$.



(a) (10pts) What is the sheet resistance?

(b) (10pts) What is the resistance?

(c) (20pts) What is the average drift velocity of a hole in (cm/s)?

(d) (20 pts) For a hole, what is the average time between collisions (the average scattering time)?

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10. One end of a bar of semiconductor uniformly doped with $N_D = 10^{16} \text{ cm}^{-3}$ is illuminated such that the hole concentration at $x=0$ is maintained at $10^{12} / \text{cm}^{-3}$. No light penetrates beyond $x=0$. $T=300\text{K}$, $n_i = 10^{10} \text{ cm}^{-3}$, $\mu_n = 1154 \text{ cm}^2/\text{Vs}$, $\mu_p = 384.6 \text{ cm}^2/\text{Vs}$, and $\tau_n = \tau_p = 0.1 \text{ }\mu\text{s}$.

(a) (10pts) What is the optical generation rate, g_{op} , in units of $(\text{cm}^{-3} \text{ s}^{-1})$?

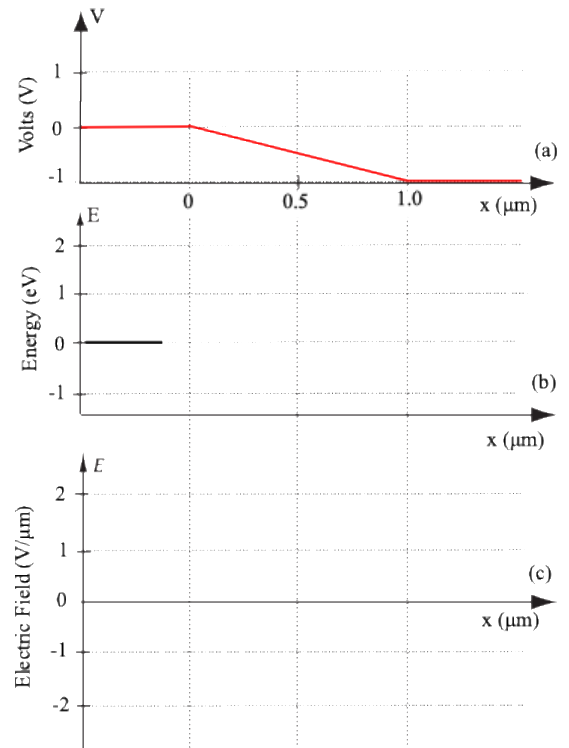
(b) (10 pts) Are the conditions of low-level injection satisfied? Why or why not?

(c) (10 pts) Calculate the electron and hole densities at $x = 100 \text{ }\mu\text{m}$.

(d) (20 pts) Calculate the hole current at $x=0$.

(e) (10 pts) If we increase the recombination times, will the hole current increase or decrease? Why?

11. A plot of the electrostatic potential is shown at right in Fig. (a).



(a) (10 pts.) On plot (b), plot the conduction band edge.
Continue from where the plot was started.

(b) (20 pts.) Plot the electric field on plot (c).

12. The band diagram of a uniformly doped semiconductor is drawn below with $V=0$.

(a) (10 pts) Sketch the band diagram with $V>0$.

(b) (10 pts) Show with an arrow next to the electrons and holes, the directions of the electron velocities, v_n , and the hole velocities, v_p .

(c) (10 pts) Show with an arrow, which way the electric field is pointing.

(d) (10 pts) Show with arrows, which way the electrical current due to electrons, J_n , is flowing.

(e) (10 pts) Show with arrows which way the electrical current due to holes, J_p , is flowing.

